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EPyT-Flow: A Toolkit for Generating Water Distribution Network Data

André Artelt [1,3*¶], **Marios S. Kyriakou** [1*], **Stelios G. Vrachimis** [1,2*], **Demetrios G. Eliades** [1*], **Barbara Hammer** [3*], and **Marios M. Polycarpou** [1,2*]

1 KIOS Research and Innovation Center of Excellence, University of Cyprus, Cyprus **2** Department of Electrical and Computer Engineering, University of Cyprus, Cyprus **3** Faculty of Technology, Bielefeld University, Germany ¶ Corresponding author * These authors contributed equally.

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Summary

This work introduces `EPyT-Flow`, an open-source Python package building on top of EPyT for facilitating water distribution network (WDN) simulations. `EPyT-Flow` provides a high-level interface for the easy generation of hydraulic and water quality scenario data. Additionally, it provides access to low-level functions of EPANET and EPANET-MSX for hydraulic and water quality modeling respectively. To accelerate research in WDN management, `EPyT-Flow` provides easy access to popular benchmark data sets for event detection and localization, and an environment for developing and testing feedback control algorithms.

Statement of need

Water Distribution Networks (WDNs) are designed to ensure a reliable supply of drinking water. These systems are operated and monitored by humans, supported by software tools, including basic control algorithms and event detectors that rely on a limited number of sensors within the WDN. These sensors measure hydraulic (e.g., pressure, flow) and water quality (e.g., chemical concentrations) states. However, given the rapid population growth of urban areas, WDNs are becoming more complex to manage due to the resulting time-varying system uncertainty. Consequently, key tasks such as event detection (e.g., leakage) and isolation, pump scheduling, and control are becoming more challenging. Moreover, modeling and predicting water quality in the distribution network is becoming more difficult due to changing environmental conditions. This is why water utilities are now driven to install even more sensors to gather data on their changing systems. Traditionally, model-based methods were used for planning and managing WDNs; however, due to rapid changes, these methods may no longer be sufficient. New AI and data-driven methods can now take advantage of big data and are promising tools for tackling challenges in water management.

Currently, non-water experts such as AI researchers face several challenges when devising practical solutions for water system applications, such as the unavailability of tools for easy scenario/data generation and easy access to benchmarks, which hinder the progress of applying AI to this domain. Easy-to-use toolboxes and access to benchmark data sets are extremely important for boosting and accelerating research, as well as for supporting reproducible research, as it was, for instance, the case in deep learning and machine learning where toolboxes such as TensorFlow and scikit-learn had a significant impact on boosting research.

State of the field

The modeling and simulation of hydraulic and water quality dynamics in water distribution networks have progressively advanced with the introduction of simulation software. Notably, `EPANET` (Rossman, 2000) and its extension `EPANET-MSX` (Shang et al., 2008) are foundational tools in this area. These are complemented by tools that make use of high-level programming languages, such as the `EPANET-MATLAB Toolkit (EMT)` (Eliades et al., 2016), the `Object-Oriented Pipe Network Analyzer (OOPNET)` (Steffelbauer & Fuchs-Hanusch, 2015), and the `EPANET-Python Toolkit (EPyT)` (Kyriakou et al., 2023). Moreover, `viswaternet` provides visualizations of static and time-varying attributes of EPANET-based WDNs (Thomas et al., 2023). These tools are instrumental in facilitating research in WDN resilience and response to various operational challenges.

These tools, however, do not support the creation of realistic (benchmark) scenarios with essential aspects such as realistic fault models (of leakages and sensor faults), various sensor configurations, custom control modules, and other events such as changes in water quality caused by external factors. A first step towards such software for scenario creation is the `Water Network Tool for Resilience (WNTR)` (Klise et al., 2017), which facilitates the simulation of hydraulic dynamics, and in addition, it allows the simulation of various events such as pipe breaks, disasters such as earthquakes, power outages, and fires. However, it currently does not support quality dynamics and also misses other crucial modules such as sensor configurations, and considerations of industrial controls.

The transition to Python-based open-source software (Kyriakou et al., 2023),(Klise et al., 2017), underscores a broader trend towards open-source, community-driven development in the water industry, aligning with the need for transparency, reproducibility, and innovation in the sector.

Functionality

Our Python toolbox `EPyT-Flow` provides a high-level interface for the easy generation of WDN scenario data, but also provides access to low-level functions for maximum flexibility as needed by domain experts – by this we aim to satisfy different needs and make it a toolbox for "everybody". In addition, its design and object-orientated implementation makes it easy to customize existing functionalities and implement new ones. `EPyT-Flow` builds upon `EPyT` which itself provides a Python interface to `EPANET` and `EPANET-MSX` – see Figure 1 for an illustration.

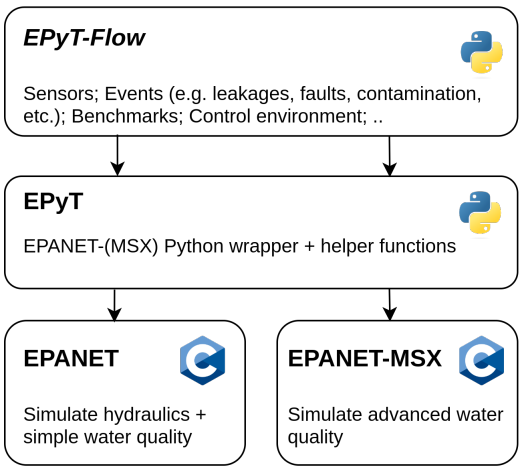

Figure 1: Illustration of the functionality of the proposed toolbox *`EPyT-Flow`*.

The toolbox currently includes 16 Water Distribution Networks (WDNs) that can be used for

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scenario generation. It goes beyond pure scenario generation by providing access to 7 popular and widely adopted benchmarks on event detection and localization (including their evaluation metrics) – ready to be utilized for building and evaluating algorithms. Furthermore, it provides an environment, inspired by the modular and extensible design of OpenAI Gym for developing and implementing control algorithms for tasks such as energy efficient pump scheduling.

To support modeling of a wide variety of scenarios, the toolbox comes with 4 different event types and a total number of 13 pre-defined and implemented events ready to be utilized in custom scenarios: 3 different leakage types, 3 actuator events, 5 sensor fault types, and 2 communication events. All the events can be easily customized by the user.

Since the measured quantities in the real world are always subject to uncertainty, the toolbox comes with 11 pre-defined types of uncertainties ranging from classic Gaussian noise to different types of (very) deep uncertainties that can be applied to hydraulic parameters such as pipe length, diameter, and roughness, water quality parameters, such as reaction coefficients, and sensor noise.

More information can be found in the comprehensive (200+ pages) EPyT-Flow Documentation and 14 fully working examples that demonstrate how `EPyT-Flow` can be utilized in different tasks.

Conclusions

In this work, we introduced a Python toolbox called `EPyT-Flow` for realistic scenario data generation and access to benchmarks of WDNs, that researchers can utilize to develop methods to support human WDN operators in various real-world challenges.

Our long-term vision for this toolbox is to split it into three parts to further facilitate the progress of research in this area:

1) A core part for data generation (i.e., scenario simulation).
2) A `BenchmarkHub` as a platform for accessing and sharing WDN benchmarks.
3) A `ModelHub` as a platform for accessing and sharing AI & classic models and algorithms for different tasks in WDNs.

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